

PROJECT SHOWCASE 02 — FULL STACK

AIRBNB CLONE

Full-Stack Property Booking Application

Developer	Madhura Surve
Architecture	Full-Stack MERN (MongoDB, Express, React, Node)
Focus Area	Session Authentication & Database Schema Workflows
Status	Completed Project / Learning Milestone
Source Code	github.com/madhurasurve708-prog/airbnb-clone

1. Project Overview

This project is a full-stack property booking application built to mimic the core features of Airbnb. The goal of this implementation was to master database-driven web application architecture, session management, API route design, and dynamic component styling. The application allows users to sign up, authenticate, browse available property listings, view detail pages, and execute reservation workflows.

2. MERN Stack Architecture

The application is structured into decoupled client and server layers. The client is a single-page application built in React that makes asynchronous API requests to a Node.js/Express.js backend server. The backend processes queries, implements validation, and coordinates transactions with a MongoDB database using an Object-Document Mapper (ODM).

Architecture Focus: Decoupling the front-end layout from the backend routing allowed me to focus on building strict RESTful controllers. By handling session authentication and protected API endpoints, the application ensures that only registered users can book properties or create custom listings.

3. Technical Stack Table

MERN Layer	Technologies	Core Role / Integration Details
Database	MongoDB, Mongoose	Storing listing parameters, booking dates, user profiles, and indexing schemas for listings search.
Backend Server	Node.js, Express.js	Configuring RESTful controllers, routing, handling CORS, and implementing session authentication.
Frontend View	React, JavaScript	Creating dynamic state-driven layouts, custom hooks for fetching data, and responsive client pages.
Styling	Vanilla CSS	Writing clean, custom stylesheets without relying on external UI libraries to master CSS Grid, Flexbox, and responsive design.

4. Key Implementation Features

- **Property Listings Registry:** Database-driven catalogs mapping property names, geographical details, pricing metrics, and host accounts.
- **Custom Search & Filters:** Query-driven routes retrieving filtered listings based on user search parameters.
- **Secure Session Authentication:** Protected route middlewares and secure password hashing ensuring safe login and booking validation.
- **Relational Booking Workflows:** Booking records linking listing IDs to user IDs with check-in/check-out date conflict validations.
- **Responsive Layout:** Custom vanilla CSS stylesheets utilizing Flexbox and Media Queries for consistent layouts across mobile, tablet, and desktop viewports.

5. Developer Role & Core Contributions

As the lead full-stack developer on this project, my work spanned both backend operations and front-end user experience. I designed the MongoDB schemas using Mongoose to ensure quick queries and data consistency. I implemented secure authentication, protected Express API endpoints, and built dynamic React views that interact with the database backend.

6. Project Learning Outcomes

Key Takeaway: Developing this Airbnb Clone provided hands-on experience with asynchronous state management, API integration, and database relationship mapping in MongoDB. Creating a responsive, premium UI with vanilla CSS reinforced best practices in responsive design, Flexbox layouts, and browser compatibility.